

SKILL-6

To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
dhars@DHARSHAN: /mnt/c/L
Microsoft Windows [Version 10.0.26300.9550]
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C:\Users\dhars>wsl
dhars@DHARSHAN: /mnt/c/Users/dhars$ cd ossp
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ nano skill6.c
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ gcc skill6.c -o skill6
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ ls
skill6 skill5.c skill6 skill6.c
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ ./skill6
===== SKILL 6: ESCAPE CHARACTERS AND PROCESS EXECUTION =====

1. Escape Character Demonstration
Normal text: Hello World
Escaped space: Hello\ World
Escaped double quote: \"Hello\"
Escaped backslash: C:\\Users\\Dharshan
Special character: \$

2. Creating Child Process
Parent Process Started
Parent PID: 680
Waiting for child process...
Child Process Started
Child PID: 681
Parent PID: 680

3. Executing Program Using exec()
total 40
-rwxrwxrwx 1 dhars dhars 16000 Oct  6 13:59 skill5
-rwxrwxrwx 1 dhars dhars 1147 Oct  6 13:59 skill5.c
-rwxrwxrwx 1 dhars dhars 16344 Oct  6 14:06 skill6
-rwxrwxrwx 1 dhars dhars 1642 Oct  6 14:06 skill6.c

Child process completed.
Child exit status: 0
```

```
===== SKILL 6 COMPLETED =====
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ pid_t = fork();
-bash: syntax error near unexpected token `('
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ nano testfork.c
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ gcc testfork.c -o testfork
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ ./testfork
Parent process
Parent PID: 732
Child PID: 733
Child process
Child PID: 733
Parent PID: 732
Child process completed
dhars@DHARSHAN: /mnt/c/Users/dhars/ossps$ |
```